NGUYEN Duc Hoan

CONTACT INFORMATION

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RESEARCH INTERESTS

Kernel Methods, Statistical Learning Theory, Regularization Methods, Machine Learning

EMPLOYMENT

- Postdoctoral at CRAN, Nancy, France — Dec 2024 - present
- Postdoctoral at RICAM, Austrian Academy of Sciences, Austria — Jan - Nov 2024
- Ph.D. candidate at RICAM, Austrian Academy of Sciences, Austria — 2020 - 2023
- Reseacher in Artificial Intelligence Laboratory, Thang Long University — 2019 - 2020
- Lecturer in Department of Mathematics and Informatics, Thang Long University — 2018 - 2020

EDUCATION

- Ph.D. at Johannes Kepler University, Linz, Austria — Sep 2020 - Nov 2023
  - Supervisors: Prof. Sergei Pereverzyev, Prof. Bernhard A. Moser, and Dr. Werner Zellinger
  - Thesis: Regularization in Reproducing Kernel Hilbert Spaces for Covariate Shift Domain Adaptation
- Master ACSYON at University of Limoges, Limoges, France — 2016 - 2017
  Master ACSYON: Algorithmics, Symbolic Computation and Numerical Optimization
  - Advisors: Prof. Jean-Guy Caputo and Prof. Arnaud Knippel.
  - Master Thesis: Inverse source problem in a forced wave graph.
- Master 1 in Hanoi Institute of Mathematics, Vietnam — 2015 - 2016
  International Master Program in Mathematics
- Bachelor of Mathematics in Hanoi University of Science, Vietnam — 2010 - 2014

PUBLICATIONS

- **D. H. Nguyen**, R. Borsoi, S. Miron, and D. Brie. "Adaptation de domaine non supervisée par pondération d'importance pour des données tensorielles", *GRETSI*, 2025.
- **D. H. Nguyen**, S. Pereverzyev, and W. Zellinger. "General regularization in covariate shift adaptation". *Data-driven Models in Inverse Problems*, edited by Tatiana A. Bubba, Berlin, Boston: De Gruyter, 2025, pp. 245-270. https://doi.org/10.1515/9783111251233-007
- **D. H. Nguyen**, W. Zellinger, and S. Pereverzyev. "On regularized Radon-Nikodym differentiation". *Journal of Machine Learning Research*, 25(266):1–24, 2024.
- M.-C. Dinu, M. Holzleitner, M. Beck, **D. H. Nguyen**, A. Huber, H. Eghbal-zadeh, B. Moser, S. V. Pereverzyev, S. Hochreiter, and W. Zellinger. "Addressing parameter choice issues in unsupervised domain adaptation by aggregation". *In: International Conference on Learning Representations (ICLR)*, selected as **notable-top-5% paper**, 2023.

- E. R. Gizewski, L. Mayer, B. A. Moser, **D. H. Nguyen**[1], S. Pereverzyev Jr, S. V. Pereverzyev, N. Shepeleva, and W. Zellinger. "On a regularization of unsupervised domain adaptation in RKHS". *Applied and Computational Harmonic Analysis*, 57:201–227, 2022.
- W. Zellinger, N. Shepeleva, M. Dinu, H. Eghbal zadeh, **D. H. Nguyen**, B. Nessler, S. Pereverzyev, and B. A. Moser. "The balancing principle for parameter choice in distance-regularized domain adaptation". *Advances in Neural Information Processing Systems*, 2021.

PREPRINTS

- Elke R. Gizewski, Shuai Lu, Stephanie Mangesius, **Hoan D. Nguyen**, Sergiy Pereverzyev Jr. *The impact of smoothness of kernels and target functions on unsupervised covariate shift adaptation in RKHS*. Submitted to Applied and Computational Harmonic Analysis, 2025. Available at https://ssrn.com/abstract=5269411

RESEARCH EXPERIENCES

- Reviewer for conferences: NeurIPS 2023, 2025, ICML 2024, 2025, and AAAI 2025.
- Work on the skin fungal diseases detection project 2019 - 2020
  Torus Company, Toulouse, France and Artificial Intelligence Lab, Thang Long University
  - Collecting images of fungal diseases, processing data, and constructing classification models
- Work on the Hanoi Formal Abstract project 2018 - 2020
  University of Pittsburgh, Carnegie Mellon University, and Thang Long University
  - Formalizing theorems of "top 100" of mathematical theorems in Lean
- Internship in Hanoi Institute of Mathematics, Vietnam Oct, 2017 - Dec, 2018
  - Advisor: Prof. Dinh Nho Hao.
  - Subject: Inverse source problem.
- Internship in INSA, Rouen, France Mar - Aug, 2017
  - Advisor: Prof. Jean-Guy Caputo and Prof. Arnaud Knippel.
  - Subject: Inverse source problem in a forced wave graph.

TEACHING EXPERIENCES

- Teaching Assistant: Data science for insurance and finance, Fall semesters 2025
  University of Lorraine, France.
- Exercise session: Mathematics for AI, Summer and Winter semesters in 2022, 2023
  Johannes Kepler University, Austria.
- Exercise session: Discrete Mathematics, Spring and Fall semesters 2019
  Thang Long University, Vietnam.

AWARDS AND FELLOWSHIPS

- Master scholarship, LabEX Sigma Lim, University of Limoges, France. 2016 - 2017
- Annual Scholarship for excellent students, Vietnam National University. 2012 - 2014

SKILLS

- Languages: Vietnamese (Native), English (Fluent), French (A1), German (A1)
- Computer skills
  - Software: MATLAB, PyTorch, TensorFlow
  - Programming: C/C++, Python, Lean

[1]Corresponding author.

REFERENCES

- ⋆ Prof. Dr. Sergei Pereverzyev
  Johann Radon Institute for Computational and Applied Mathematics
  Austrian Academy of Sciences
  Email: sergei.pereverzyev@oeaw.ac.at

- ⋆ Prof. Jean-Guy Caputo
  Laboratoire of Mathematiques
  INSA de Rouen
  Email: caputo@insa-rouen.fr

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  Hanoi Institute of Mathematics
  Vietnam Academy of Science and Technology
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